
Erasure: A Stated Goal Reorganises What a Language Model Deletes

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Abstract

1 A generator asked to write imports everything it knows. A generator allowed only
2 to delete cannot import anything, so every choice it makes is a choice about the
3 words already in front of it. We hold that subtractive operator fixed, change one
4 word of the instruction, and measure what changes. Told the remainder should
5 be a summary, Claude Opus 5 protects proper nouns harder than any other word
6 class (survival log-odds +0.70). Told the remainder should be a poem, it cuts
7 them harder than any other word class (−0.73), against +0.03 for matched random
8 deletion: a swing of 1.44 log-odds produced by one word. Both selections are
9 internally stable and mutually distinct, agreeing with themselves at Jaccard 0.541
10 and 0.699 but with each other at 0.233, against a chance floor of 0.138. Across
11 four models from two vendors, the reorganisation itself is general and the direction
12 is not: gpt-oss-120b *keeps* proper nouns for a poem (+0.50) where Opus cuts them.
13 Deleting the names is a property of one model family, not of language models. We
14 release the corpus, 3,800 verified erasures, and an exhibit that shades every source
15 word by how often it survived.

16 1 Constraint as an instrument for reading agency

17 Ask a language model to write a poem about a statute and you cannot tell where the poem came
18 from. The output is drawn from everything in the weights, and the source text is one influence among
19 millions. Creative agency in that setting is real but illegible: distributed across a training corpus, a
20 decoding procedure, and a prompt, with no way to attribute any particular word to any particular
21 cause.

22 Take away the ability to add. Require the output to be a subsequence of the input, in order, with
23 nothing invented. The search space collapses from all of language to the power set of one passage.
24 Every word present is a word the model chose to keep, every word absent one it chose to cut, and
25 both are checkable against the source. Subtraction does not make the model more creative. It makes
26 its taste *auditable*.

27 This is the sense in which the work addresses agency. The constraint is a refusal built into the
28 interface: the system is denied its most characteristic capacity, generation, and what remains is
29 selection. Erasure poets have used this refusal as a compositional method for fifty years, from
30 Johnson’s *Radi os* over *Paradise Lost* to Phillips’ *A Humument*. What has not been done is to run the
31 operator under controlled conditions and measure the selection function it exposes.

32 We do that, and vary one thing. Two instructions differ in a single clause naming what the remainder
33 should be: *a poem*, or *a summary of the passage*. Everything else, operator, corpus, sampling, and
34 validation, is identical. Whatever separates the two outputs is attributable to the stated goal and to
35 nothing else.

36 Contributions.

- 37 • A controlled measurement of how far a model’s selection diverges when only the stated goal
38 changes, against a chance floor and matched random deletion (Section 3.1).
- 39 • A reversal in which proper nouns move 1.44 log-odds between goals: protected by the
40 summary, targeted by the poem (Section 3.2).
- 41 • A cross-vendor arm showing which half generalises. The reorganisation does; the direction
42 does not (Section 3.5).
- 43 • A corpus of 100 freely licensed passages in five registers, 3,800 alignment-verified erasures
44 with failures retained, and an exhibit of the selection function as a gradient over the source.

45 2 The operator, and the guarantee

46 Each attempt sends one passage and one instruction. The rules state that deletion is the only permitted
47 action, that kept words must stay in original order, that punctuation and capitalisation may be dropped,
48 and that the response go inside tags. The literary tradition is never named: calling this “erasure poetry”
49 would summon a genre and return a pastiche of it rather than the model’s own selection. Lineation
50 is never mentioned either, so any line breaks in the poem condition are unprompted and therefore a
51 finding rather than an instruction.

52 The constraint is checkable, so we check it rather than trust it. Every response is aligned back against
53 its source and rejected if one word cannot be found there, in order. Across 3,800 attempts, 3,336
54 are compliant; for Opus 5 alone compliance is 93.1%. Failures are kept in the record rather than
55 resampled away, because a model breaking a rule it cannot have misread is itself a result.

56 Alignment is the step that could have manufactured a finding. When a word repeats, many embeddings
57 are valid and they disagree about *where* the kept material came from, which is what positional analysis
58 measures. Greedy-leftmost matching resolves every repeated `the` to the first one and reports a front-
59 loading bias the model never had. Among valid embeddings we therefore take the one preserving the
60 most adjacency: *in the beginning was the end*, erased to *the end*, returns indices 4 and 5, not 1 and 5.

61 **Corpus.** 100 passages of 110–188 words, 20 in each of five registers, all freely licensed: statute
62 (eCFR), science (arXiv abstracts), news (Wikinews, CC BY 2.5), fiction (Project Gutenberg), and
63 poetry (the WISP public-domain set). Everything arrives as a single paragraph with no layout, poems
64 included, because lineation is a cue the other four registers lack and leaving it in would let the model
65 identify poetry by its shape. Our public-domain filter is stricter than the one WISP ships, which
66 marks a poem public domain from the original poet’s dates: right about the poem, wrong about the
67 text on the page, since a 2002 translation of Sappho is in copyright no matter when Sappho died.

68 **Conditions and baselines.** Four conditions cross two goals with two length regimes: *free*, where
69 no length is stated and retention is the model’s own answer, and *budget*, where the instruction asks
70 for 15–25% of the words so that composition is comparable between goals rather than confounded by
71 how much each keeps. Five independent samples per passage per condition. Every attempt is matched
72 by four baselines generated at that attempt’s own keep-count: random, rare-word, content-word, and
73 lead. Random deletion is the control every claim below is measured against.

74 3 Results

75 3.1 The goal changes the selection, and both selections are stable

76 Five independent erasures of the same passage under the same instruction agree at Jaccard 0.541
77 (poem) and 0.699 (summary), against a chance floor of 0.138. Across the two goals, agreement falls
78 to 0.233: above chance, so the goals share something, but only 43% of what the poem condition
79 shares with itself. The paired within-minus-across difference is +0.289 [+0.263, +0.315], Wilcoxon
80 $p = 1.1 \times 10^{-17}$, $n = 98$ (Figure 1).

81 There are two selection functions here. Each is repeatable. They want different words.

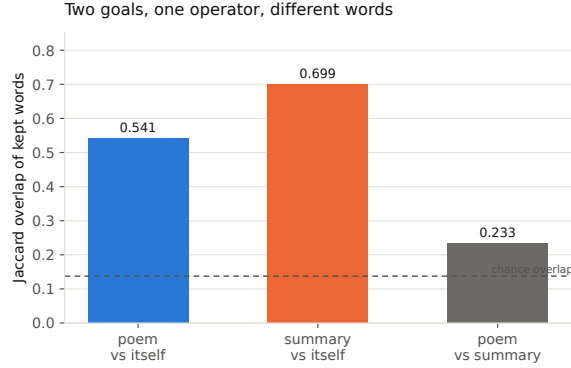


Figure 1: One operator, two goals. Each instruction reproduces itself far above chance and agrees with the other barely above it. The stable difference between the bars is what the rest of the paper decomposes.

82 3.2 The reversal: proper nouns

83 Holding length fixed at a matched 24% budget, survival log-odds by part of speech separate the
 84 two goals sharply, and one class separates them most (Figure 2). The summary keeps proper nouns
 85 at $+0.702$ [$+0.530, +0.871$]. The poem cuts them at -0.732 [$-0.928, -0.558$]. Matched random
 86 deletion sits at $+0.027$ [$-0.043, +0.093$], $p = 0.427$, exactly where a control should. The gap is
 87 $+1.435$ [$+1.205, +1.676$].

88 The summary side is the expected direction and functions as validation: entity salience is an estab-
 89 lished driver of extractive summarisation, so a positive coefficient on proper nouns confirms the
 90 harness. The unstudied half is the poem driving the same class to -0.73 , and the swing between
 91 them.

92 The words that fall to zero survival are the ones that say which thing this is: wikinews,
 93 president, north, 42, pursuant, administrative, emission. The words that sur-
 94 vive best are home (83%), wildlife (80%), shadow (68%), water (64%), hope (60%). Kept
 95 words in the poem condition run $+0.156$ [$+0.123, +0.187$] more concrete than their source, against
 96 -0.005 for random deletion at the same length. Verbs, nouns and adjectives survive above the
 97 passage rate; proper nouns, numerals and subordinating conjunctions below it. Both goals pull from
 98 the opening: mean kept position 0.41 (poem) and 0.40 (summary) against 0.50 for random deletion.

99 The operation is visible in a single passage. From a Wikinews report (CC BY 2.5), under the two
 100 instructions at matched length:

101 **Poem.** arrived by plane / beginning his visit / turn the world’s attention to Africa / peace,
 102 migration, / dialogue / the continent / among the quickest growing / more than one-fifth / of
 103 the world’s total

104 **Summary.** Pope Leo XIV arrived in Algeria beginning visit to Africa; the eleven-day visit
 105 would address peace, migration, and interfaith dialogue. also includes stops in Cameroon,
 106 Angola, and Equatorial Guinea.

107 The poem has deleted the pope, the country, the broadcaster, and three of the four nations. The
 108 summary has kept all of them. Same passage, same operator, same budget, one word of difference in
 109 the instruction.

110 3.3 Told to make a poem, the model barely cuts

111 With no length stated, the summary keeps 41.1% of the words and the poem keeps 57.5% ($n = 95$
 112 paired passages, $p = 6.3 \times 10^{-15}$). The poetic operation is the *less* selective one. Erasure as a form is
 113 built on deep cuts, and asked for a poem with no budget the model declines to make them. Whatever
 114 it has learned “poem” to mean, it is not the tradition of the sparse page.

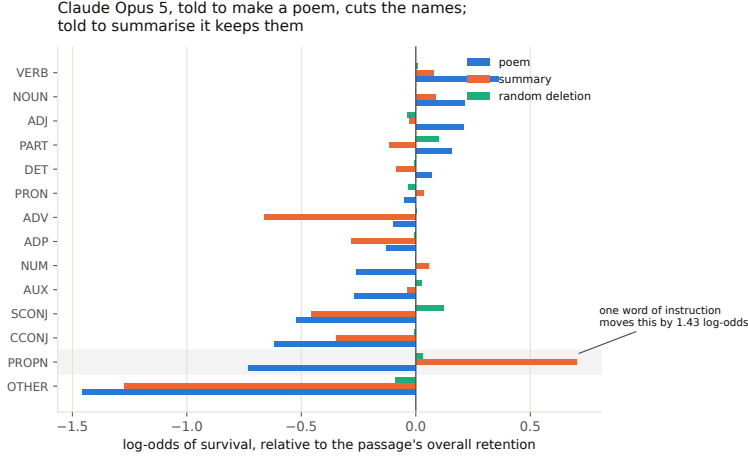


Figure 2: Survival log-odds by part of speech, relative to each passage’s own retention. Proper nouns are the class the goal moves furthest, and matched random deletion sits at zero throughout.

Table 1: Cluster bootstrap over passages ($B = 2000$), 95% intervals. Differences are paired within replicate. Per-condition separation intervals are omitted: resampling duplicates passages, which shrinks within-register spread and inflates a ratio that divides by it.

Quantity	Estimate	95% CI	p
PROPN log-odds, poem	-0.732	[-0.928, -0.558]	<0.001
PROPN log-odds, summary	0.702	[0.530, 0.871]	<0.001
PROPN log-odds, random deletion	0.027	[-0.043, 0.093]	0.427
PROPN gap, summary minus poem	1.435	[1.205, 1.676]	<0.001
separation, poem_budget minus random deletion	-0.078	[-0.140, -0.001]	0.042
separation, summary_budget minus random deletion	0.172	[0.082, 0.251]	<0.001
concreteness shift, poem	0.156	[0.123, 0.187]	<0.001
concreteness shift, random deletion	-0.005	[-0.018, 0.008]	0.460
concreteness shift, poem minus random	0.161	[0.124, 0.197]	<0.001
within-goal minus across-goal Jaccard	0.289	[0.263, 0.315]	1.1×10^{-17}

115 3.4 What survives: register

116 We expected erasure to delete what a text sounds like while keeping what it is about. That prediction
 117 failed twice, in direction and in strength. Style-based register accuracy is 59.3% for poem erasures
 118 against 64.1% for matched random deletion, which does not separate at $n = 100$.

119 The underlying geometry carries more. Separation, between-register centroid distance over within-
 120 register spread, runs 1.108 for the source, 0.916 for the summary, 0.749 for random deletion and
 121 0.667 for the poem. Passages are the unit of replication, so comparisons use a cluster bootstrap over
 122 passages with differences paired inside each replicate. The summary holds registers apart better than
 123 chance deletion, comfortably, at +0.172 [+0.082, +0.251]. The poem pulls them together, but the
 124 interval’s upper end sits at -0.001 with $p = 0.042$, which is where a stricter design would break it.
 125 We report it as suggestive and build nothing on it. Per-condition intervals are deliberately omitted:
 126 resampling duplicates passages, duplicates shrink within-register spread, and a ratio dividing by that
 127 spread is inflated, while the paired differences are not.

128 3.5 Four models: three skills that come apart

129 Obeying delete-only, obeying a length budget, and cutting the names are separate capacities, and
 130 they order the models differently (Figure 3, Table 2). Opus 5 respects the subsequence constraint
 131 on 97% of budgeted erasures, gpt-oss-120b on 91%, Sonnet 5 on 78%, Haiku 4.5 on 77%. Told to
 132 keep 15–25%, Opus keeps 24%, Sonnet 25% and gpt-oss 23%, while Haiku keeps 58% and does not

Three separate skills, and only one of them crosses the vendor line

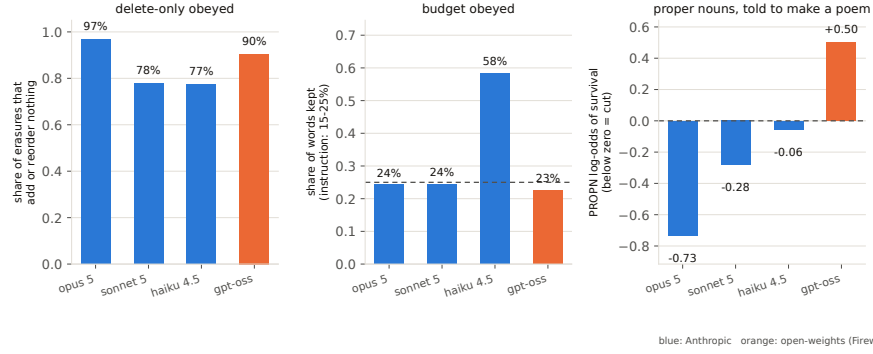


Figure 3: Three capacities, measured separately across four models and two vendors. Only the middle panel orders the models by scale; only the right panel crosses the vendor line, and it crosses it with the sign flipped.

Table 2: Across models. Compliance is the share of erasures adding or reordering nothing. Retention is measured against an instruction to keep 15–25%.

Model	Poem compliance	Summary compliance	Retention	PROPN log-odds
opus-5	96.2%	97.4%	24.2%	-0.73
haiku-4-5	80.7%	74.0%	58.3%	-0.06
sonnet-5	94.3%	61.3%	24.5%	-0.28
gpt-oss-120b	94.7%	86.3%	22.6%	0.50

appear to register the instruction as binding, which is a documented weakness of instruction-based length control rather than a discovery. Since the three orderings disagree, the three skills have to be treated separately.

The cross-vendor arm changed what this paper can claim, and it splits into two statements that must not be merged.

The reorganisation is general. Both models keep proper nouns harder for a summary than for a poem, each by an interval clear of zero, and both agree with themselves more than with the other goal (Wilcoxon $p = 1.1 \times 10^{-17}$ for Opus, 4.4×10^{-7} for gpt-oss). A stated purpose reorganises what a model is willing to delete.

Which words go is not general. Opus cuts proper nouns to make a poem at -0.73 $[-0.93, -0.56]$. gpt-oss keeps them at $+0.50$ $[+0.36, +0.65]$. The intervals are nowhere near each other. Opus’s goal gap is $+1.44$ $[+1.21, +1.68]$ against gpt-oss’s $+0.42$ $[+0.24, +0.62]$, roughly three times larger. Within the Anthropic family the effect fades down the capability ladder: -0.73 for Opus 5, -0.28 for Sonnet 5, -0.06 for Haiku 4.5. Deleting the names is a signature of one family, not a property of language models, and any sentence claiming otherwise is falsifiable in an afternoon by a reviewer with an API key.

Sonnet’s summary condition is the informative failure: compliant on only 61.3% of attempts, with 83 of 139 violations being invented words, usually a small inflection the syntax wanted, cited bent into citing. Summarising under a hard delete-only rule is harder than writing a summary.

4 The exhibit

A single erasure shows what the model kept once. It cannot show what the model wanted. A word kept in five of five attempts and a word kept in one of five are both simply present in one output.

The exhibit prints fifteen passages, three per register, each shown whole with every word shaded by how often it survived across five erasures, beside one of those erasures as returned. The gradient is the selection function itself, rendered as a property of the source rather than of any single output: the

158 parts of a statute a model finds unspeakable in a poem are visible as pale text before any erasure is
159 chosen.

160 From 36 CFR Part 1, at 23% kept:

161 the proper government of persons
162 and natural and cultural resources
163
164 will conserve scenery
165 provide enjoyment
166 leave them unimpaired
167 for future generations

168 Nothing here evaluates whether these are good poems. The exhibit is the argument for that question,
169 and it is an argument for a reader to settle.

170 5 Related work

171 Deletion-only rewriting where the output must be a token subsequence of the input is **sentence**
172 **compression by deletion**, going back to Filippova et al. [1]. The summary condition here *is* that
173 task. A reviewer will say this is compression with a poetry prompt, and the reply belongs in the
174 introduction rather than the rebuttal: compression is the control, not the confound, and the result
175 is the distance between it and the poem under an identical operator. Length-control failure under
176 instruction is documented [2], and entity salience driving extractive content is established [3]; both
177 appear here in the expected direction, which is evidence the pipeline reproduces effects it should.

178 Computational erasure poetry exists as artist tools [4, 5], none of which measures what is preserved,
179 uses a controlled corpus, or compares against matched random deletion. Erasure as critical method
180 has a literary-critical and legal literature [6, 7], a different genre of claim from a measurement.

181 Searching four ways for a quantitative characterisation of what *human* erasure poets delete, with the
182 final pass restricted to ACL Anthology, arXiv, ACM, Springer, Oxford Academic and the digital-
183 humanities abstracts index, returned criticism and craft guides only. We claim we are not aware of
184 such a characterisation, not that none exists.

185 6 Limitations

186 The human baseline is missing, and it is the comparison this study most obviously wants. It cannot
187 be assembled from published work: the three canonical erasures with recoverable public-domain
188 sources, *Radi os*, *A Humument* and Bervin’s *Nets*, are themselves in copyright, so kept-word sets
189 cannot be released. It has to come from annotators. The instrument is built and driven end to end: a
190 self-contained offline page where clicking a word toggles it, so an annotator physically cannot add or
191 reorder anything, with the on-screen instruction generated from the same string the models are sent.
192 The data is not collected.

193 The style classifier is trained on 100 texts and cannot resolve differences below roughly six points,
194 which is why the register claim rests on the separation ratio instead. Concreteness norms cover
195 91.9% of source words and the valence, arousal and dominance norms 48.0%, so affective shifts are
196 estimated on about half the vocabulary. Cross-model part-of-speech figures are read at each model’s
197 own retention, and Haiku’s 58% is a different regime from Opus’s 24%.

198 Two bugs were caught by reading failures rather than by tests, and both would have entered the
199 record as findings. A normaliser folding em dashes to hyphens welded `shall--the` into one token,
200 logging a model performing exactly the operation under study as having invented a word, and it
201 misfired systematically on whoever punctuates with dashes. Baselines seeded with Python’s salted
202 `hash()` made two passes over identical data disagree by four points on the random control.

203 **Conclusion.** Denying a generative model the ability to generate does not silence it. It makes its
204 preferences readable. A stated purpose reorganises what a system will destroy, across vendors; that
205 making a poem means deleting the names belongs to one model family and should be named as such.

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